Early silicification, not oxygenation, governed the preservation of microfossil morphology across the Great Oxidation Event

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Abstract

The diversification of morphologically diagnostic microfossils after the Great Oxidation Event (GOE; $\sim$2.4 Ga) is widely read as a biological signal, yet low atmospheric oxygen before the GOE should, in principle, have slowed the oxidative degradation of organic matter and favoured preservation. Compilation of published petrographic, spectroscopic and isotopic data for chert-hosted biotas on either side of the GOE shows that morphological fidelity tracks the timing and completeness of early silicification rather than any measure of bottom-water oxygenation. Within the 1.88 Ga Gunflint biota, morphospecies that preserve internal cellular contents carry 10^8–10^9 intracellular Fe atoms μm^{-3}, whereas empty sheaths remain at the matrix background ($\sim 10^6$), documenting a within-assemblage covariation of cellular fidelity with early mineralization. The silica cycle did not change across the GOE: Precambrian seawater was saturated with respect to amorphous silica throughout, and silica was not drawn down until the much later radiation of biosilicifiers. Neither does the sulfur isotope record support a clean redox control on preservation; the sulfate rise it registers began in the Neoarchean and bears on the decay of organic matter rather than on the silicification that gates cellular fidelity. The biomarker evidence once taken to place eukaryotes and cyanobacteria before the GOE is now regarded as contamination. Morphological richness across the GOE is therefore most parsimoniously interpreted as a facies signal conditioned by silicification, and the post-GOE "rise" of microfossils cannot be read as a biological radiation without accounting for this preservational structure.

Keywords: Great Oxidation Event; taphonomy; silicification; Archaean–Proterozoic; microfossils; iron speciation; molecular clocks.

1 Introduction

It has long been recognised that anoxia should retard the degradation of sedimentary organic matter, and that the destruction of cellular morphology during early decay proceeds more slowly where molecular oxygen is scarce (Canfield, 1994; Hedges & Keil, 1995). From this premise one might expect the Archaean ocean, in which dissolved oxygen was negligible (Pavlov & Kasting, 2002) and sulfate scarce (Habicht et al., 2002; Crowe et al., 2014), to have been a favourable setting for the preservation of organic-walled microfossils. The rock record appears to say the opposite. Morphologically simple, mat-related carbonaceous remains are present locally in Archaean successions, but assemblages of taxonomically diagnostic microfossils become abundant and widespread only after the Great Oxidation Event (Javaux & Lepot, 2018; Wacey et al., 2014). The contrast is exemplified by the gunflintiacean biotas of the $\sim$1.88 Ga Gunflint Iron Formation, which preserve filaments, spheroids and spindle-shaped forms in cellular detail (Lepot et al., 2017), against the kerogenous globules and rare filaments of the $\sim$2.72 Ga Tumbiana Formation (Lepot et al., 2009).

Two interpretations of this pattern are routinely advanced. The first is biological: oxygenic photosynthesis, or the eukaryotic cell, radiated after the GOE, and the record documents that diversification. The second is preservational: the conditions required to preserve morphology became more widely available, so that pre-existing diversity simply became visible. The two are not mutually exclusive, but the distinction matters. If the latter contributes substantially, first-appearance data cannot be taken at face value as a record of origination.

This paper assembles the relevant published petrographic, spectroscopic and isotopic evidence for chert-hosted biotas spanning the GOE and asks what governs the preservation of cellular morphology across that interval. The synthesis points to early silicification, rather than oxygenation, as the decisive variable, and carries direct consequences for how the early fossil record is read.

2 Background

2.1 Redox across the GOE

The disappearance of mass-independent sulfur isotope fractionation (S-MIF) between ~2.45 and 2.32 Ga fixes the GOE as the interval in which atmospheric O_2 rose irreversibly above $\sim 10^{-5}$ of the present atmospheric level (Holland, 2006; Bekker et al., 2004; Pavlov & Kasting, 2002; Farquhar et al., 2000). The transition was protracted and regionally diachronous (Lyons et al., 2014; Philippot et al., 2018). Iron speciation, the principal local redox proxy, indicates that anoxic ferruginous conditions dominated deposition throughout the Archaean and much of the Proterozoic (Canfield, 1998; Poulton & Canfield, 2005, 2011), euxinia being spatially restricted (Reinhard et al., 2009). Seawater sulfate rose across the GOE from micromolar to millimolar levels (Habicht et al., 2002; Kah et al., 2004; Crowe et al., 2014), but the sulfur isotope evidence for this rise extends into the Neoarchean (Reinhard et al., 2009), and trace-metal and sulfur isotopic signals of oxidative weathering are now known to predate the GOE in several basins (Anbar et al., 2007).

2.2 The microfossil record

The Archaean record of cellular morphology is sparse and contested. The filamentous structures described from the ~3.46 Ga Apex chert (Schopf, 1993) were reinterpreted as abiogenic (Brasier et al., 2002), and although better-preserved Archaean microbiotas are known from the ~3.4 Ga Strelley Pool Formation (Sugitani et al., 2010; Delarue et al., 2020), they remain modest in comparison with the post-GOE assemblages. Gunflint-type biotas, with their distinctive filaments and spheroids, are widespread in shallow-marine cherts of ~1.9 Ga age and permit taxonomic discrimination (Javaux & Lepot, 2018). The post-GOE increase in the diversity and abundance of morphologically recognisable microfossils is the pattern that invites explanation.

2.3 The silica cycle

Before the evolution of silica-biomineralising organisms, the ocean was saturated or supersaturated with respect to amorphous silica. Silica concentrations of order ~2 mM, two to three orders of magnitude above the modern value, are inferred for Archaean and Proterozoic seawater, with abiological precipitation the dominant sink (Siever, 1992; Perry & Lefticariu, 2007); the role of early silicification in exceptional preservation extends into the Neoproterozoic (Tarhan et al., 2016). The drawdown of seawater silica was driven by the much later radiation of sponges, radiolaria and ultimately diatoms, and is essentially a Neoproterozoic to Phanerozoic phenomenon. The capacity for early silica precipitation was therefore available on both sides of the GOE.

3 Material and approach

Six chert-hosted biotas were compared (Table 1): the Strelley Pool Formation, the Moodies Group and the Tumbiana Formation (pre-GOE), and the Belcher Group, the Gunflint Iron Formation and the Sokoman Iron Formation (post-GOE). These units span shallow-marine to peri-tidal silicified facies of comparable, low metamorphic grade, allowing the influence of thermal maturity to be held approximately constant. For each, published descriptions of petrography, Raman spectroscopy of carbonaceous material, iron mineralogy and sulfur geochemistry were compiled; no new measurements are reported. Where published constraints are absent, that is stated rather than inferred.

Table 1: Preservation of cellular morphology in six chert-hosted biotas across the GOE. Grades are qualitative summaries of the cited literature.

Unit (age)	GOE	Facies / maturity	Preservation of morphology	Silicification
Strelley Pool (∼3.4 Ga)	pre	chert; Raman ∼300°C	diverse (lenticular, spinose, "tail-like") but of lower cellular fidelity than Gunflint; walls partly silica-replaced	early
Moodies Gp (∼3.22 Ga)	pre	siliciclastic tidal; low grade	tufted and cavity-dwelling mats; cellular detail poorly resolved	weak
Tumbiana Fm (∼2.72 Ga)	pre	stromatolitic carbonate/chert; <300°C	abundant kerogenous globules; rare filaments; no cellular assemblage	weak
Belcher Gp (∼2.0 Ga)	post	chert-replaced stromatolites; low grade	diverse microbiota, chert-permineralised	early
Gunflint Fm (∼1.88 Ga)	post	chert; prehnite–pumpellyite	diverse, high cellular fidelity; intracellular Fe-silicate/carbonate	early
Sokoman IF (∼1.88 Ga)	post	chert; low grade	Gunflint-type assemblage	early

Sources: (Sugitani et al., 2010; Delarue et al., 2020; Wacey et al., 2012; Homann et al., 2015; Lepot et al., 2009; Decreane et al., 2021; Lepot et al., 2017; Wacey et al., 2013; Alleon et al., 2016; Javaux & Lepot, 2018).

The strongest single line of evidence is drawn from nanoscale petrography of the Gunflint biota (Lepot et al., 2017), in which the relationship between cellular preservation and early mineralization can be assessed quantitatively and within a single assemblage, avoiding the facies-averaging that limits between-formation comparison.

4 The control on morphological fidelity

4.1 Within-assemblage evidence from Gunflint

At the scale of individual microfossil populations, the Gunflint biota shows that the preservation of internal cellular structure is coupled to early mineralization. Morphospecies retaining cellular contents, including thick-walled *Huroniospora* and Type-2 *Gunflintia*, carry between 3×10^8 and 10^9 intracellular Fe atoms $\mu\mathrm{m}^{-3}$, together with finer intra-microfossil quartz than the surrounding matrix (Lepot et al., 2017). Empty sheaths, in contrast, including Type-1 *Gunflintia minuta*, *Animikiea* and thin-walled *Huroniospora*, are depleted in iron to the level of the chert matrix background ($\sim 1.5 \times 10^6$ atoms $\mu\mathrm{m}^{-3}$; Fig. 1). Whether the intracellular iron reflects in vivo biomineralization or very early diagenetic precipitation, its restriction to morphospecies with preserved contents indicates that cellular fidelity required mineral entombment before decay had destroyed internal structure.

A focused-ion-beam comparison of the Strelley Pool and Gunflint biotas reached a compatible conclusion at the between-assemblage scale (Wacey et al., 2012): although the Strelley Pool microfossils are morphologically diverse, their cellular fidelity is lower than that of Gunflint, with wall organic matter partly replaced by silica. Morphological diversity and cellular fidelity are therefore not the same quantity, and both are graded with the timing and completeness of silicification.

4.2 Between-assemblage pattern

Across the six units, the preservation of cellular morphology tracks the quality of early silicification, independently of position relative to the GOE (Table 1). The pre-GOE Strelley Pool biota, where silicification was early and complete, preserves a diverse morphological assemblage, whereas the pre-GOE Tumbiana and Moodies biotas, in which silicification was weak or confined to mat textures,

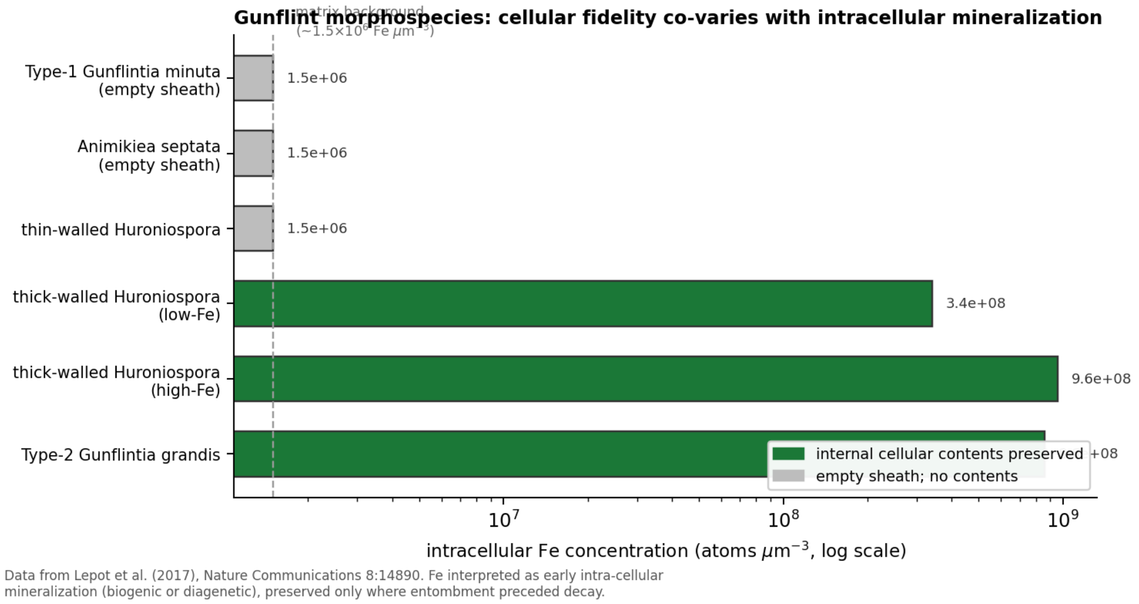

Figure 1: Intracellular Fe concentration in Gunflint morphospecies, against the preservation of cellular contents. Morphospecies with preserved internal structure carry 10^8–10^9 Fe atoms μm^{-3}; empty sheaths remain at the matrix background. Data from Lepot et al. (2017).

preserve kerogen and mats rather than cells. The post-GOE Gunflint, Sokoman and Belcher biotas are all early-silicified and morphologically rich. Pyrite, where present, is a poor indicator of fidelity: in Gunflint, pyritisation is confined to patches, develops under locally sulfate-limited conditions, and destroys rather than preserves organic structure (Wacey et al., 2013; Lepot et al., 2017). Pyritisation is also documented in pre-GOE Tumbiana stromatolites (Decreane et al., 2021), so it cannot mark a post-GOE preservational threshold.

4.3 Why redox and the silica cycle cannot supply a GOE control

If a change in redox across the GOE had governed morphological preservation, the proxy record of that change should align with the palaeontological pattern. It does not. Iron speciation indicates pervasive ferruginous deposition on both sides of the GOE (Poulton & Canfield, 2011), and the sulfur isotope signal of rising sulfate extends back into the Neoarchean (Reinhard et al., 2009; Philippot et al., 2018). The sulfate rise registered by δ^{34}S bears on microbial sulfate reduction and the sulfurisation of organic matter, a pathway that affects the survival of bulk kerogen rather than the templating of cellular morphology. The relevant control on morphology, the availability of early silica, was unchanging across the GOE: Precambrian seawater remained silica-saturated throughout, and the silica cycle was not reorganised until the much later appearance of biosilicifiers (Siever, 1992; Perry & Lefticariu, 2007).

5 Origins, first appearances, and the limits of inference

If the post-GOE increase in morphological diversity reflects, even in part, the unmasking of lineages that existed but were not previously preserved, then independent temporal evidence should place those lineages before the GOE. Molecular clock analyses now place the stem origin of cyanobacteria, and with it oxygenic photosynthesis, in the Archaean, at ∼3.3–3.6 Ga (Sánchez-Baracaldo, 2021), whereas the crown-group diversification of extant Oxyphotobacteria is estimated to postdate the rise of oxygen (Shih et al., 2017). Molecular clock estimates for the last eukaryotic common ancestor remain model-dependent and scattered, but several precede the first widely accepted eukaryotic microfossils, leaving a protracted gap between molecular and palaeontological estimates of divergence

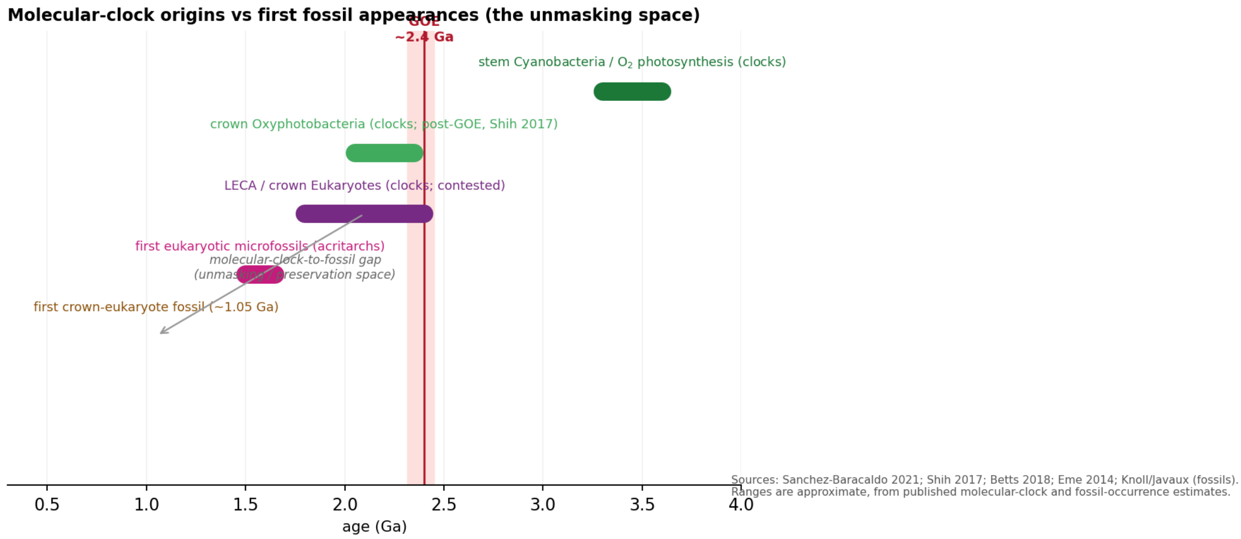

Figure 2: Molecular clock estimates of lineage origin against first fossil appearances. The stem origin of cyanobacteria lies in the Archaean (Sánchez-Baracaldo, 2021); crown Oxyphotobacteria diversify around or after the GOE (Shih et al., 2017); eukaryote molecular-clock estimates precede the first widely accepted eukaryotic fossils (Betts et al., 2018; Eme et al., 2014). Ranges are approximate.

(Betts et al., 2018; Eme et al., 2014) (Fig. 2).

The biomarker record cannot at present resolve this gap. The steranes and 2-methylhopanes reported from ~2.7 Ga shales, once taken as evidence for Archaean eukaryotes and cyanobacteria (Brocks et al., 1999), were shown to reside in younger carbonate veins (Rasmussen et al., 2008), and analysis of ultraclean drill cores has demonstrated that hopane and sterane concentrations in Archaean rocks are indistinguishable from procedural blanks (French et al., 2015). The molecular fossil evidence for pre-GOE eukaryotes and oxygenic photosynthesis must therefore be regarded as contaminant.

The consequence is a logical bind. First-appearance data are themselves the preservational signal under examination, and the biomarker evidence that might have provided an independent check is contaminated. The question of whether the post-GOE pattern reflects evolution or unmasking cannot be settled from the rock record alone, but the uncertainty runs in one direction: molecular clocks are consistent with lineages that predate the GOE, and a facies control on preservation is sufficient to account for their late appearance. Reading the post-GOE diversification as a biological radiation, in the absence of correction for preservation, is not warranted.

6 Discussion

The compiled evidence supports a single, consistent reading. The preservation of cellular morphology in Precambrian cherts is conditioned by early silicification, a facies and early-diagenetic variable that operated throughout the Archaean and Proterozoic but was expressed unevenly, depending on the silica budget of the depositional environment and the timing of cementation relative to decay. Where silica entombment was early and complete, as at Gunflint and, to a lesser degree, Strelley Pool, morphology survives in cellular detail; where it was weak or absent, as at Tumbiana and Moodies, only kerogen and mat textures remain (Fig. 3). Oxygenation, and the sulfate rise that accompanied it, modified the decay of organic matter and the sulfurisation of kerogen (Lepot et al., 2009; Decreane et al., 2021) but did not gate the silicification on which cellular fidelity depends.

This reading carries several implications. First, abundance and diversity counts compiled across the GOE cannot be interpreted as proxies for biomass or evolutionary innovation without correction for the silicification available in the sampled facies; the well-exposed and intensively sampled gunflint-type cherts of the Palaeoproterozoic are liable to over-represent the biology of their time relative to

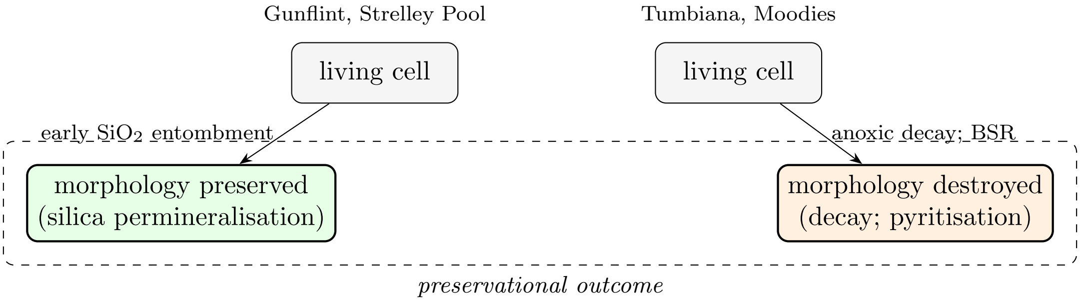

Figure 3: Preservational pathways. Where early silica permineralisation outpaces decay, cellular morphology is retained; where decay (and pyritisation) precedes or accompanies mineralization, morphology is lost and only kerogen survives. The balance is set by depositional facies and the timing of cementation, not by the oxygenation state of ambient water.

less completely silicified Archaean successions. Second, the gap between molecular clock estimates of origination and the first appearance of fossils, most conspicuously for eukaryotes (Betts et al., 2018), is consistent with a preservational lag and need not denote a late evolutionary origin. Third, for the search for biosignatures on Mars and in other ancient successions, the relevant variable for the preservation of morphology is a silica-precipitating facies, not the presence of oxygen.

The principal limitation is that the comparison rests on a synthesis of published petrographic descriptions rather than a uniform new dataset. The within-Gunflint relationship between intracellular iron and cellular fidelity (Lepot et al., 2017) is quantitative, but the between-formation pattern is semi-quantitative and would be strengthened by a standardised, per-population scoring of silicification and fidelity measured under identical conditions on matched samples. A regression of morphological fidelity on a quantitative silicification index, within single formations of comparable thermal maturity, would convert the inference advanced here into a direct test, and would do so with greater power than any comparison couched as a two-group contrast across the GOE.

7 Conclusions

Across the GOE, the preservation of microfossil morphology in Precambrian cherts is governed by early silicification. Within the Gunflint biota, cellular fidelity co-varies quantitatively with intracellular mineralization; between biotas, fidelity tracks the quality of silicification and is independent of position relative to the GOE. The silica cycle did not change across the GOE, the sulfur isotope evidence for the sulfate rise extends into the Neoarchean and bears on organic-matter decay rather than on morphological templating, and the biomarker evidence for pre-GOE eukaryotes and cyanobacteria is best regarded as contaminant. The post-GOE diversification of morphologically diagnostic microfossils is therefore most parsimoniously read as a facies signal, conditioned by silicification, and should not be interpreted as a biological radiation without preservational correction. A within-formation, quantitative test of fidelity against silicification is the appropriate next step.

References

Alleon, J., Bernard, S., Le Guerróué, N., et al. (2016). Molecular preservation of 1.88 Ga Gunflint organic microfossils as a function of temperature and mineralogy. *Nature Communications* 7, 11977.

Anbar, A. D., Duan, Y., Lyons, T. W., et al. (2007). A whiff of oxygen before the Great Oxidation Event? *Science* 317, 1903–1906.

Bekker, A., Holland, H. D., Wang, P.-L., et al. (2004). Dating the rise of atmospheric oxygen. *Nature* 427, 117–120.

Betts, H. C., Puttick, M. N., Clark, J. W., et al. (2018). Integrated genomic and fossil evidence illuminates life's early evolution and eukaryote origin. *Nature Ecology & Evolution* 2, 1556–1562.

Brasier, M. D., Green, O. R., Jephcoat, A. P., et al. (2002). Questioning the evidence for Earth's oldest fossils. *Nature* 416, 76–81.

Brocks, J. J., Logan, G. A., Buick, R., & Summons, R. E. (1999). Archean molecular fossils and the early rise of eukaryotes. *Science* 285, 1033–1036.

Canfield, D. E. (1994). Factors influencing organic carbon preservation in marine sediments. *Chemical Geology* 114, 315–329.

Canfield, D. E. (1998). A new model for Proterozoic ocean chemistry. *Nature* 396, 450–453.

Crowe, S. A., Døssing, L. N., Beukes, N. J., et al. (2014). Sulfate was a trace constituent of the Archean ocean. *Science* 346, 735–739.

Decreane, S., Philippot, P., Ader, M., et al. (2021). Intense biogeochemical iron cycling revealed in Neoarchean Tumbiana Formation stromatolites. *Geochimica et Cosmochimica Acta*.

Delarue, F., Robert, F., Derenne, S., et al. (2020). Out of rock: a new look at the morphological and geochemical preservation of microfossils from the 3.46 Gyr-old Strelley Pool Formation. *Precambrian Research* 336, 105472.

Eme, L., Sharpe, S. C., Brown, M. W., & Roger, A. J. (2014). On the age of eukaryotes: evaluating evidence from fossils and molecular clocks. *Cold Spring Harbor Perspectives in Biology* 6, a016139.

Farquhar, J., Bao, H., & Thiemens, M. (2000). Atmospheric influence of Earth's earliest sulfur cycle. *Science* 289, 756–758.

French, K. L., Hallmann, C., Hope, J. M., et al. (2015). Reappraisal of hydrocarbon biomarkers in Archean rocks. *Proceedings of the National Academy of Sciences* 112, 5915–5920.

Habicht, K. S., Gade, M., Thamdrup, B., Berg, P., & Canfield, D. E. (2002). Calibration of sulfate levels in the Archean ocean. *Science* 298, 2372–2374.

Hedges, J. I., & Keil, R. G. (1995). Sedimentary organic matter preservation: an assessment and speculative synthesis. *Marine Chemistry* 49, 81–115.

Holland, H. D. (2006). The oxygenation of the atmosphere and oceans. *Philosophical Transactions of the Royal Society B* 361, 903–915.

Homann, M., Heubeck, C., Airo, A., & Tice, M. M. (2015). Morphological adaptations of 3.22 Ga-old tufted microbial mats to Archean coastal–habitat (Moodies Group, Barberton Greenstone Belt). *Precambrian Research* 266, 47–64.

Javaux, E. J., & Lepot, K. (2018). The Paleoproterozoic fossil record: implications for the evolution of the biosphere during Earth's middle-age. *Earth-Science Reviews* 176, 68–94.

Kah, L. C., Lyons, T. W., & Frank, T. D. (2004). Low marine sulfate and protracted oxygenation of the Proterozoic biosphere. *Nature* 431, 834–838.

Lepot, K., Benzerara, K., Rividi, N., et al. (2009). Organic matter heterogeneities in 2.72 Ga stromatolites: alteration versus preservation by sulfur incorporation. *Geochimica et Cosmochimica Acta* 73, 6579–6599.

Lepot, K., Addad, A., Knoll, A. H., et al. (2017). Iron minerals within specific microfossil morphospecies of the 1.88 Ga Gunflint Formation. *Nature Communications* 8, 14890.

Lyons, T. W., Reinhard, C. T., & Planavsky, N. J. (2014). The rise of oxygen in Earth's early ocean and atmosphere. *Nature* 506, 307–315.

Pavlov, A. A., & Kasting, J. F. (2002). Mass-independent fractionation of sulfur isotopes in Archean sediments: strong evidence for an anoxic Archean atmosphere. *Astrobiology* 2, 27–41.

Perry, E. C., & Lefticariu, P. (2007). Formation and geochemistry of Precambrian cherts. In *Treatise on Geochemistry* (2nd ed.), vol. 7, 99–113.

Philippot, P., Ávila, J. N., Killingsworth, B. A., et al. (2018). Globally asynchronous sulphur isotope signals require re-definition of the Great Oxidation Event. *Nature Communications* 9, 2245.

Poulton, S. W., & Canfield, D. E. (2005). Development of a sequential extraction procedure for iron: implications for iron partitioning in continentally derived particulates. *Chemical Geology* 214, 209–221.

Poulton, S. W., & Canfield, D. E. (2011). Ferruginous conditions: a dominant feature of the ocean through Earth's history. *Elements* 7, 107–112.

Rasmussen, B., Fletcher, I. R., Brocks, J. J., & Kilburn, M. R. (2008). Reassessing the first appearance of eukaryotes and cyanobacteria by 2-Ga-old hydrocarbon biomarkers. *Nature* 455, 1101–1104.

Reinhard, C. T., Raiswell, R., Scott, C., Anbar, A. D., & Lyons, T. W. (2009). A late Archean sulfidic sea stimulated by early oxidative weathering of the continents. *Science* 326, 713–716.

Sánchez-Baracaldo, P. (2021). The Archean origin of oxygenic photosynthesis and extant cyanobacterial lineages. *Proceedings of the Royal Society B* 288, 20210675.

Schopf, J. W. (1993). Microfossils of the early Archean Apex chert: new evidence of the antiquity of life. *Science* 260, 640–646.

Shih, P. M., Hemp, J., Ward, L. M., Matzke, N. J., & Fischer, W. W. (2017). Crown group Oxyphotobacteria postdate the rise of oxygen. *Geobiology* 15, 19–29.

Siever, R. (1992). The silica cycle in the Precambrian. *Geochimica et Cosmochimica Acta* 56, 3265–3272.

Sugitani, K., Lepot, K., Nagaoka, T., et al. (2010). Biogenicity of morphologically diverse carbonaceous microstructures from the ca. 3400 Ma Strelley Pool Formation, in the Pilbara Craton, Western Australia. *Astrobiology* 10, 899–920.

Tarhan, L. G., Hood, A. v. S., Droser, M. L., Gehling, J. G., & Briggs, D. E. G. (2016). Exceptional preservation of soft-bodied Ediacara Biota promoted by silica-rich oceans. *Geology* 44, 951–954.

Wacey, D., Menon, S., Green, L., et al. (2012). Taphonomy of very ancient microfossils from the ∼3400 Ma Strelley Pool Formation and ∼1900 Ma Gunflint Formation: new insights using a focused ion beam. *Precambrian Research* 220–221, 234–250.

Wacey, D., McLoughlin, N., Kilburn, M. R., et al. (2013). Nanoscale analysis of pyritized microfossils reveals differential heterotrophic consumption in the ∼1.9 Ga Gunflint chert. *Proceedings of the National Academy of Sciences* 110, 8020–8024.

Wacey, D., Saunders, M., & Brasier, M. D. (2014). Changing the picture of Earth's earliest fossils (3.5–1.9 Ga) with new approaches and discoveries. *Proceedings of the National Academy of Sciences* 111, 8778.